

Types and Properties of Solutions

1. Solution

2. Dilute Solution

3. Raoult's law

4. Types of solution, units of Concentration

mole fraction.

5. Colligative Properties.

6. Molecular Weight from Raoult's law

7. Ideal and real solution

8. Deviation

9. Henry's law and its application

10. Nernst distribution law.

⇒ Solution: A solution is a homogeneous mixture of two or more substances on molecular level.

→ The constituent of the mixture present in a smaller amount is called the solute and the one present in a larger amount is called the solvent.

⇒ Dilute solution: If the concentration of the solute in a solution is very low, the solution is called a dilute solution.

⇒ Concentrated solution: If the concentration of the solute in a solution is high or remain in greater amount, then the solution is called a concentrated solution.

⇒ Types of solutions:

	<u>State of solute</u>	<u>State of solvent</u>	<u>Example</u>
1.	Gas	Gas	Air
2.	Gas	Liquid	Oxygen in water, CO ₂ in water (drinks)
3.	Gas	Solid	Absorption of H ₂ by Palladium
4.	Liquid	Liquid	Alcohol in water
5.	Liquid	Solid	Mercury in silver
6.	Solid	Liquid	Sugar, Salt
7.	Solid	Solid	Carbon in iron (steel)

⇒ Concentration of a solution:

The concentration of a solution is defined as:
'the amount of solute present in a given amount of solution.'

$$\text{So, Concentration} = \frac{\text{Quantity of solute}}{\text{Volume of solution}}$$

⇒ Different ways of expressing concentration:

i) Per cent by weight (% by W)

ii) Mole fraction (x)

iii) Molarity (M)

iv) Normality (N)

v) Molality

vi) ppm, ppb, ppt.

→ Per cent by weight: It is the weight of the solute as a per cent of the total weight of the solution.

$$\% \text{ by weight of solute} = \frac{\text{Wt. of solute}}{\text{Wt. of solution}} \times 100$$

→ Mole Fraction: The ratio of the number of moles of solute and the total number of moles of solute and solvent is called mole fraction.

$$x_{\text{solute}} = \frac{n}{n+N}$$

$$x_{\text{solvent}} = \frac{N}{n+N}$$

$$x_{\text{solute}} + x_{\text{solvent}} = 1$$

n = moles of solute

N = moles of solvent

x = Mole fraction

→ Molarity: Molarity is defined as the number of moles of solute per litre of solution.

$$\therefore \text{Molarity (M)} = \frac{\text{moles of solute (n)}}{\text{Volume in litres (V)}}$$

→ Molality: Molality of a solution is defined as the number of moles of solute per kilogram of solvent.

$$\text{Molality (m)} = \frac{\text{moles of solute}}{\text{mass of solvent in Kg}}$$

→ Normality (N): Normality of a solution is defined as number of gram equivalent of solute per litre of the solution.

$$\text{Normality (N)} = \frac{\text{Number of gm equivalent of solute}}{\text{Vol}^m \text{ of sol}^n \text{ in litre}}$$

$$[\text{gram equivalent weight} = \frac{\text{Amount of given substance}}{\text{Molecular weight}}]$$

(Gram equivalent weight of 5g NaCl)

$$= \frac{5}{58.5} = 0.0854$$

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Lecture Sheet

PROPERTIES OF DILUTE SOLUTIONS.

Q.1. What is meant by Additive & Constitutive property?

Additive property:-

An additive property is defined as the property, the aggregate value of which is the sum of values of its constituents.

Constitutive property:-

A constitutive property is defined as the property which depends upon the constitution or structure of the molecule like arrangement of atoms within the molecule, multiple bonds & rings etc.

Q.2. What are colligative properties? [2002, 2003].

OR. Name the colligative properties of dilute solutions. How can you prove that elevation of boiling point is a colligative property? [92, (5)].

ANSWER: The word 'colligative' is derived from Latin word "colligatus" that means "to bind together or going together". The essential features of colligative properties is that they depend only on the number of solute particle present in solution. Being closely related to each other through a common explanation, these have been grouped together under the class name colligative properties.

Therefore, those properties of the solutions which depend only upon the total number of molecules of solute per unit volume & not on its chemical nature are called colligative properties.

Four main colligative properties of solutions are as follows:

1. Lowering of vapour pressure
2. Elevation of boiling point
3. Depression of freezing point
4. Osmotic pressure.

The above four colligative properties of solutions are so intimately connected with each other that if one property is measured, the other can be calculated. For this reason, these properties are called colligative properties.

The colligative properties are widely used for the determination of molecular mass of substances.

⇒ Elevation of Boiling Point:

When a liquid is heated, its vapour pressure rises and when it equals the atmospheric pressure, the liquid boils. The addition of a non-volatile solute lowers the vapour pressure and consequently elevates the boiling points as the solution has to be heated to a higher temperature to make its vapour pressure become equal to atmospheric pressure. If

T_b is the boiling point of the solvent and T is the boiling point of the solⁿ, the difference in the boiling points (ΔT) is called the elevation of boiling point.

$$\therefore T - T_b = \Delta T$$

⇒ Freezing point Depression:

→ The freezing point is the temperature at which a liquid turns into solid. The freezing point of water is 0° Celsius.

→ The freezing point depression is the difference in temperature between the freezing point of the pure solvent and that of the solution.

If T_f be the freezing point of pure solvent and T_1 be the freezing point of the solution, then ΔT is called the freezing point depression.

$$\text{i.e., } T_f - T_1 = \Delta T.$$

⇒ Osmosis and Osmotic Pressure:

- Osmosis: The flow of the solvent through a semipermeable membrane from pure solvent to solution or from a dilute solution to concentrated solution, is termed as Osmosis. (Greek Osmos = to push)
- Osmotic pressure: Osmotic pressure may be defined as the external pressure applied to the solution in order to stop the osmosis of solvent into solution separated by a semipermeable membrane.
- It is a colligative properties because it depends on the number of solute particles in a solution and it's directly proportional to the number of moles of solute in a solution.

✓ Elevation of boiling point is a colligative property —

The thermodynamic expression of elevation of boiling point is given as below:

$$\Delta T_b = \frac{RT_b^2}{L_e} \cdot x_2$$

$$\Delta T_b = \frac{RT_b^2}{L_e} \cdot \frac{n_2}{n_1} \quad \left[\text{For a dilute solution, } x_2 \approx \frac{n_2}{n_1} \right]$$

From the above equation, it is clear that the elevation of boiling point depends only on the number of moles of solute in a dilute solution but does not depend on its nature. Thus, elevation of boiling point is a colligative property.

*** Q.3 Show that, the lowering of vapour pressure is a colligative property.
 ANSWER:

According to the Raoult's law for lowering of vapour pressure — The relative lowering of vapour pressure of a solvent by a non-volatile solute is equal to the mole fraction of the solute in solution. Mathematically it may be written,

$$\frac{p_0 - p}{p_0} = \frac{n_2}{n_1 + n_2}$$

From the above equation, it is clear that the lowering of vapour pressure depends only on the mole number of solute in a solution but independent of its nature. Thus, the lowering of vapour pressure is a colligative property.

p^0 = Vapour pressure of solvent
 p = Vapour pressure of solution
 n_1 = mole number of solvent
 n_2 = mole number of solute.

⇒ State and explain Raoult's Law
and deduce an expression for Raoult's law:

Raoult's Law: The relative lowering of the vapour pressure of a dilute solution is equal to the mole fraction of the solute present in dilute solution.

Raoult's law can be expressed mathematically in the form:

$$\frac{P_0 - P_s}{P_0} = \frac{n}{n+N}$$

Where, P_0 & P_s are the vapour pressure of the solvent and solution respectively.

And, n & N are the number of moles of solute and solvent respectively.

⇒ Derivation of Raoult's Law:

The vapour pressure of the pure solvent is caused by the number of molecules evaporating from its surface. When a non-volatile solute is dissolved in solⁿ, the pressure of solute molecules in the surface blocks a fraction of the surface, where no evaporation takes place.

This causes the lowering of the vapour pressure.

The vapour pressure of the solution is therefore, determined by the number of molecules of the solvent present at any time in the surface which is proportional to the mole fraction.

That is, $p_s \propto \frac{N}{n+N}$

Where, N = moles of solvent and
 n = moles of solute.

$$\text{OR, } P_s = K \frac{N}{n+N} \quad (1)$$

K being proportionality factor,

In case of pure solvent, $n=0$ and hence

$$\begin{aligned} \text{mole fraction of solvent} &= \frac{N}{n+N} \\ &= \frac{N}{0+N} = 1 \end{aligned}$$

Now, from eqⁿ (1), the vapour pressure, $P=K$

Therefore, the eqⁿ (1) assumes the form:

$$P_s = P \frac{N}{n+N}$$

$$\Rightarrow \frac{P_s}{P} = \frac{N}{n+N}$$

$$\Rightarrow 1 - \frac{P_s}{P} = 1 - \frac{N}{n+N}$$

$$\therefore \frac{P - P_s}{P} = \frac{n}{n+N}$$

This is Raoult's Law.

Q. 5 What are the causes of deviations of colligative properties in a solution from ideal behaviour?

ANSWER: There are the following causes that deviate the colligative properties of a solution from ideal behaviour are given below:

01. When the concentration of the solution is high, i.e., the solution is concentrated.
02. When the solute in the solution is associated.
03. When the solute in the solution suffers electrolytic dissociation.
04. When the solution is non-metal. (volatile)

Q. 6 When are the laws of colligative properties valid?

ANSWER:- The laws on colligative properties are applicable/valid in the following cases are given below:

- 1). When the solution is dilute.
- 2). When the solute is non-volatile & non-electrolytes.
- 3). When the solute in the solution does not undergo molecular association & dissociation.
- 4) When the solution is metal.

✓ Q.9 Determine the molecular weight from Raoult's law of lowering of vapour pressure.

ANSWER: We can measure molecular weight of a dissolved substance from Raoult's law of lowering of vapour pressure. We find from Raoult's law -

$$\frac{p^0 - p}{p^0} = \frac{n}{N+n} = x_2$$

For a dilute solution $\frac{n}{N+n}$ in above equation may be replaced by $\frac{n}{N}$. as n is negligible in comparison to N . The equation may be written as -

$$\frac{p^0 - p}{p^0} = \frac{n}{N}$$

If w_2 gm of solute of molecular weight M_2 be present in w_1 gm of solvent of molecular weight M_1 , then.

$$\frac{n}{N} = \frac{w_2/M_2}{w_1/M_1}$$

And the equation becomes.

$$\frac{p^0 - p}{p^0} = \frac{w_2/M_2}{w_1/M_1}$$

By measuring p^0 & p for a solution of known concentration, M_2 may be obtained.

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Q. What are ideal solutions? What are the thermodynamic criteria of such solutions? [2000, (2+4)]

OR, What are ideal & real solutions? Explain causes of non-ideality of solution. [98, (2+3=5)]

OR, What is an ideal solution? What types of binary systems yield ideal solutions? [95, (4)]

Short notes → 2002, 2001.

Tutorial

Ideal solution: — / Ideality of solⁿ: —

The solution which obeys Raoult's law at any temp. and pressure is called ideal solution.

When ideal solution is prepared —

1. It does not absorb heat
2. It does not evolve heat
3. No volume change occurs.

Thermodynamic criteria of ideal solution: —

1. Volume of mixing, $\Delta V_{mix} = 0$
2. Enthalpy of mixing, $\Delta H_{mix} = 0$
3. Entropy of mixing, $\Delta S_{mix} = 0 - \sum x_i R \ln x_i$

When the ideal solution is prepared by two compounds A & B the intermolecular forces between A-A & B-B and A-B must be same. Example — Hexane-Heptane, Benzene-toluene

Non-ideal solution: — / Non ideality of solⁿ: —

A solution which shows deviation from Raoult's law is called non-ideal solution. Following are the

main reason why a given solution may not behave ideally —

1. If the concentration of the solution be high.
2. If the solute is associated or dissociated in solution.
3. In the solution if the internal pressure be different.
4. If compound is formed between solvent & solute

If does not obey Raoult's law or the interactions of A and B molecules in the solution are not similar to those of pure A & pure B or $\Delta V_{mix} \neq 0$ & $\Delta H_{mix} \neq 0$

Types & Properties of Solutions

Lecture Sheet

Q. 10 State & explain Henry's law with application & limitation.
OR, State Henry's law. 1998 [2]

OR Short note $\rightarrow$ Henry's law [2002]
or, Short note $\rightarrow$ Henry's law & its application [1995]

Statement:—

The influence of pressure on the solubility of a gas was expressed by N. Henry. His conclusions, generally known as Henry's law:—

The mass of gas dissolved by a given volume of solvent, at constant temperature is proportional to the pressure of the gas with which it is in equilibrium.

It is stated mathematically.

$$W \propto P$$

$$\Rightarrow W = KP$$

Where,

$W \rightarrow$ mass of the gas dissolved in a unit volume of the solvent.

$P \rightarrow$ Pressure of the gas in equilibrium with the solvent

$K \rightarrow$ Constant, called Henry's law constant, characteristic of the nature of the gas, the nature of the solvent & the temperature.

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Another statement of Henry's law:—

The volume of the gas dissolved, measured at the pressure used, is independent of the pressure.

Like distribution law, Henry's law holds good for dilute solutions of gases which does not react with solvent. If a mixture of gases is in contact with a liquid, the partial pressure of an individual gas, not their total pressure determine the mass of each gas dissolving. In such cases, the solubility of each gas is proportional to its partial pressure.

■ Application:—

The influence of partial pressure on solubility is utilized in making carbonated beverages like beer, Champagne & many soft drinks. So called "Soda water" is bottled under a carbon dioxide pressure of about 4 atm. When the bottle is opened to the air, the partial pressure of CO_2 above the solution decreased about (0.05 atm) & bubbles out.

Limitation & Validity of Henry's law:—

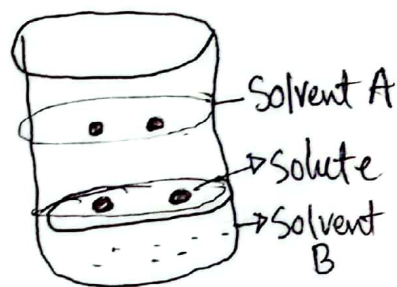
- ① Henry's law is obeyed fairly satisfactorily by many gases of low solubility, provided the pressure is not too high or the temperature is not too low.
- ② It holds good for dilute solutions of gases which do not react with the solvent.
- ③ This law is strictly applicable to those gases where the molecular species are the same in the liquid phase.

⇒ Nernst Distribution Law:

If we take two immiscible solvents A and B in a beaker, they form separate layers. When a solute 'X' which is soluble in both solvents is added, it gets distributed between them.

Molecules of 'X' will pass from solvent A to B and from solvent B to A. Finally a dynamic equilibrium is set up.

At equilibrium, the rate at which molecules of 'X' pass from one solvent to the other is balanced.



WH Nernst a German Chemist gave a law for distribution of solute in two immiscible solvents. In 1891, he studied the distribution of several solutes between different appropriate pairs of solvents.

He gave a generalization which governs the distribution of a solute betⁿ two non-miscible solvents.

Statement of Nernst's Distribution Law:

If a solute X distributes itself between two immiscible solvents 'A' and 'B' at constant temperature and 'X' is in the same molecular condition in both solvents.

$$\frac{\text{Concentration of X in A}}{\text{Concentration of X in B}} = K_D$$

If C_1 denotes the conc. of the solute in solvent A and C_2 denotes the conc. of solute in solvent B,

Then Nernst's Distribution law can be expressed mathematically as:

$$\frac{C_1}{C_2} = K_D$$

K_D or K is called 'Distribution coefficient' or 'Partition coefficient'.

Application of Distribution Law: —

1. The distribution law has been successfully applied to the study of association or dissociation of a solute in a solvent.
2. It can be used to determine the equilibrium constant K_c for a complex reaction. ($KI + I_2 \rightleftharpoons KI_3$)
3. It finds numerous applications in the process of extraction of a solute from solvent both in the laboratory & in the industry.
4. It can be used to determine the solubilities of a solute.
5. It can be used to confirmatory test for bromide & iodide.
6. It can be applied to purify the lubricating oil.

Limitation of Distribution Law: —

1. The temperature is kept constant throughout the experiment.
2. The solution must be dilute.
3. The two solvents must be immiscible.
4. The concentration of the solute are noted after the equilibrium has been reached.